***Genetics, Teratology, and Antenatal Diagnosis*: Professional Examination**

1. A 26-year-old pregnant woman is exposed to a known environmental agent that alters biochemical pathways during organogenesis, resulting in a structural birth defect. An agent that causes such permanent structural anomalies is defined as a:
a. Mutagen
b. Carcinogen
c. Pathogen
d. Mitogen
e. Teratogen
2. During which specific phase of human intra-uterine development is the embryo most highly susceptible to gross structural malformations from teratogenic exposures?
a. Blastocyst phase (Days 1–5)
b. Implantation phase (Days 6–12)
c. Fetal period (Months 3–9)
d. Embryonic period (Weeks 3–8)
e. Pre-embryonic period (Weeks 1–2)
3. An infant is born with missing long bones of the limbs, where the hands and feet are attached directly to the trunk (phocomelia). This specific birth defect is classically linked to maternal ingestion of which drug?
a. Tetracycline
b. Phenytoin
c. Alcohol
d. Thalidomide
e. Warfarin
4. A newborn presents with a microcellular chromosomal deletion leading to a congenital anomaly. Structural or functional birth defects that are evident immediately at the time of delivery are formally classified as:
a. Idiopathic anomalies
b. Phenocopies
c. Teratogenic syndromes
d. Metabolic errors
e. Congenital anomalies
5. A structural defect resulting from an intrinsically abnormal developmental program, typically caused by genetic factors or inherited mutant genes, is defined as a:
a. Disruption
b. Deformation
c. Syndrome
d. Malformation
e. Association
6. An amniocentesis is scheduled for a pregnant woman to evaluate the fetal karyotype. According to standard clinical guidelines, this invasive antenatal procedure is most safely performed during which gestational timeframe?
a. Weeks 6 to 8
b. Weeks 10 to 12

c. Weeks 15 to 20
d. Weeks 24 to 28
e. Weeks 32 to 36

7. A child is diagnosed with severe microcephaly, cardiac valvular defects, and a smooth philtrum with a thin upper lip. This constellation of signs is pathognomonic for prenatal exposure to:

a. Cigarette smoke
b. Lithium carbonate
c. Tetracycline
d. Alcohol (Ethanol)
e. Retinoic acid

8. Which non-invasive antenatal diagnostic technique is used routinely during the first and second trimesters to evaluate nuchal translucency, fetal growth, and major structural anomalies?

a. Amniocentesis
b. Chorionic villus sampling
c. Cordocentesis
d. Fetoscopy
e. Ultrasonography

9. A newborn exhibits microphthalmia, cataracts, and a patent ductus arteriosus. The mother experienced a low-grade fever and a maculopapular rash during her first trimester. Which infectious teratogen is responsible?

a. Toxoplasma gondii
b. Cytomegalovirus
c. Herpes simplex virus
d. Treponema pallidum
e. Rubella virus

10. A pregnant woman requires chronic treatment for acne. Her physician strictly avoids prescribing oral isotretinoin (Retinoic acid) because it acts as a severe teratogen by disrupting:

a. Neural crest cell migration and craniofacial development
b. Primary yolk sac constriction
c. The basement membrane of the primordial follicles
d. The tight junctions of the blood-testis barrier
e. Mitotic division of the type A spermatogonia

11. A 38-year-old pregnant woman undergoes non-invasive prenatal testing (NIPT) that suggests an abnormality. To confirm the chromosomal diagnosis early in the first trimester (Weeks 10–12), which invasive test should be selected?

a. Amniocentesis
b. Chorionic villus sampling (CVS)
c. Percutaneous umbilical blood sampling (PUBS)
d. Maternal serum alpha-fetoprotein screening
e. Fetoscopy

12. When a normal fetal structure faces mechanical compression or molding by prolonged external forces—such as uterine oligohydramnios—the resulting birth defect is classified as a:

a. Disruption
b. Deformation

c. Malformation
d. Syndrome
e. Association

13. Maternal serum screening reveals an abnormally elevated level of alpha-fetoprotein (MSAFP). This finding is strongly indicative of which underlying fetal developmental defect?

a. Down syndrome (Trisomy 21)
b. Duodenal atresia
c. Neural tube defect (e.g., Anencephaly or Spina bifida)
d. Cleft palate
e. Amelogenesis imperfecta

14. A pregnant woman with poorly controlled Type 1 Diabetes Mellitus faces a higher risk of giving birth to an infant with congenital heart defects or caudal dysgenesis. This highlights the role of:

a. Infectious teratogens
b. Maternal metabolic factors (Teratogenesis)
c. Autosomal dominant mutations
d. Heavy metal toxicities
e. X-linked recessive genes

15. If a genetic mutation or teratogenic insult causes an entire group of anomalies to occur together repeatedly due to a single, specific underlying cause, this collection of defects is termed a:

a. Disruption
b. Association
c. Sequence
d. Malformation
e. Syndrome

16. In a conceptual scenario, a teratogenic drug is ingested during the first two weeks of development (prior to gastrulation). According to the principles of teratology, what is the most likely developmental outcome?

a. Severe localized phocomelia of the upper limbs
b. An "all-or-none" effect, resulting in either early spontaneous abortion or completely normal recovery
c. Isolated cleft palate with macroglossia
d. Transposition of the great vessels
e. Underdevelopment of the mandibular process only

17. If a teratogen selectively inactivates the specific homeobox (HOX) genes responsible for craniocaudal patterning, what conceptual defect will be observed in the embryo?

a. The blood-testis barrier will mechanically degrade immediately
b. Severe segment-specific skeletal alterations and positional misalignments along the body axis
c. The zona pellucida will fail to hatch from the blastocyst
d. Complete failure of the syncytiotrophoblast to produce hCG
e. Accelerated expansion of the amniotic cavity fluid

18. A newborn is found to have a severe structural abnormality of the bowel wall caused by an intrauterine vascular accident (amniotic band entrapment) that mechanically tore a normally developing structure. This is classified as a:

a. Malformation
b. Deformation
c. Disruption
d. Syndrome
e. Association

19. Why does the timing of a teratogenic exposure dictate the exact pattern of birth defects seen in a child?

a. Because teratogens can only alter the DNA sequence of the hypoblast
b. Because different organ systems possess specific, non-overlapping critical periods of susceptibility based on their rapid cellular division rates
c. Because maternal liver enzymes only function during the first trimester
d. Because the amnion completely blocks all drugs after the fourth week
e. Because the primitive streak only responds to chemical signals during the second week

20. A 42-year-old pregnant woman wishes to screen for fetal Down syndrome non-invasively using maternal blood. This technique, which isolates small pieces of fetal genetic material from the mother's circulation, is:

a. Cordocentesis
b. Amniocentesis
c. Chorionic villus sampling
d. Cell-free fetal DNA screening (NIPT)
e. Percutaneous umbilical blood sampling

21. If a teratogen completely blocks the maternal-fetal transport of folic acid during the third and fourth weeks of gestation, what is the expected clinical pathology at birth?

a. Primary tubal ectopic pregnancy
b. Neural tube defects such as anencephaly or spina bifida cystica
c. Complete absence of the primary yolk sac vesicle
d. Precocious fusion of the lower limbs (sirenomelia)
e. Neoplastic conversion of the cytotrophoblast

22. In a scenario where an inherited gene defect impairs the regular synthesis of structural type I collagen, what type of congenital abnormality is produced across the skeletal framework?

a. An environmental disruption
b. A mechanical deformation
c. A genetic malformation
d. An unlinked association
e. A metabolic sequence

23. If a patient displays a non-random collection of anomalies (such as vertebral, anal, cardiac, tracheoesophageal, renal, and limb defects) where a single root cause cannot be scientifically identified, this group is designated as a:

a. Syndrome
b. Sequence
c. Disruption
d. Association (e.g., VACTERL)
e. Deformation

24. During an antenatal check-up, a maternal serum screening test reveals very low levels of alpha-fetoprotein (MSAFP) alongside decreased estriol and elevated hCG. This biochemical pattern is a diagnostic marker for:

a. Open neural tube defects
b. Congenital megacolon
c. Treacher Collins syndrome
d. Down syndrome (Trisomy 21)
e. Sacrococcygeal teratoma

25. Prenatal exposure to therapeutic doses of Tetracycline during the second and third trimesters is strictly contraindicated in dental and obstetric practice because it acts as a teratogen that causes:

a. Complete failure of the palate to fuse
b. Permanent discoloration of the deciduous teeth and hypoplasia of enamel
c. Severe microcephaly and cleft lip
d. Flipped limb orientations and phocomelia
e. Atresia of the external auditory meatus

26. Conceptually, the "All-or-None" period of teratogenesis during the first two weeks of development exists because the early blastomeres are:

a. Completely differentiated into distinct organ tissues
b. Highly totipotent/pluripotent, allowing remaining cells to completely compensate for damaged cells, or leading to death if too many cells die
c. Resilient to any form of chemical or physical radiation
d. Nourished exclusively by the amniotic fluid bag
e. Incapable of undergoing mitotic cleavage divisions

27. If an infant is born with congenital deafness, microcephaly, and periventricular calcifications, and the mother was exposed to an asymptomatic viral infection during pregnancy, the agent is likely:

a. Rubella virus
b. Cytomegalovirus (CMV)
c. Herpes simplex virus
d. Toxoplasma gondii
e. Treponema pallidum

28. An antenatal diagnostic procedure called cordocentesis (percutaneous umbilical blood sampling) is employed primarily when a clinician needs to:

a. Obtain a pure sample of fetal blood to evaluate severe fetal anemia or isoimmunization
b. Measure the depth of the amniotic fluid cavity
c. Extract cells exclusively from the maternal decidua
d. Verify the rupture of the oropharyngeal membrane
e. Assess the structural diameter of the connecting stalk

29. A newborn presents with a cluster of secondary defects (clubfoot and pulmonary hypoplasia) that all directly originated from a single primary factor: a chronic leak of amniotic fluid (oligohydramnios). This clinical paradigm is a:

a. Syndrome
b. Association
c. Disruption
d. Sequence (e.g., Potter sequence)
e. Malformation

30. Which of the following statements represents a major conceptual advantage of choosing Chorionic Villus Sampling (CVS) over Amniocentesis for high-risk pregnancies?

a. CVS provides a completely non-invasive screening method

b. CVS measures alpha-fetoprotein directly from the amniotic fluid
c. CVS can be performed significantly earlier in pregnancy (Weeks 10–12), providing early genetic diagnostic results
d. CVS carries zero risk of causing miscarriage or fetal trauma
e. CVS allows for direct visual exploration of the fetal face using a scope

Answer Key Table

Q 1-8	Q 9-16	Q 17-24	Q 25-30
e	e	b	b

d	**a**	**c**	**b**
d	**b**	**b**	**d**
e	**c**	**b**	**b**
d	**b**	**d**	**c**
c	**e**	**b**	**c**
d	**b**	**d**	
e	**b**	**d**	